

Chapter 5

Assessing the Application of Artificial Intelligence and Machine Learning in Detecting Misinformation and Disinformation



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Abstract Digital platforms have become an increasingly important aspect for society as a source of information. As societies' reliance on digital information continues to increase, the proliferation of misinformation and disinformation presents new challenges that must be considered. These challenges, ranging from ethical dilemmas to political manipulation, will be analysed to assess the severity of misinformation and disinformation in a digital era. While exploring the formidable obstacles and challenges facing society, this chapter will examine the untapped potential of artificial intelligence (AI) and machine learning (ML) in combating the spread of false information. By drawing on the advancements in these fields, such as natural language processing, pattern recognition and anomaly detection, this chapter will discuss their viability to combat misinformation. Through the analysis of relevant literature, this chapter will provide a holistic overview of the challenges and opportunities in this domain while drawing on relevant case study examples. As digital landscapes evolve, the need for innovative detection techniques will shed light on the advancements that serve as the defence against the increasing threat of misinformation and disinformation. Content analysis methodologies will be examined to reveal how AI interprets content and verifies factual accuracy, providing insight into the evolving strategies designed to combat misinformation. An examination of algorithmic bias and fairness will highlight the ethical imperatives, ensuring that AI models contribute to the preservation of truth without inadvertently amplifying existing biases that many ML models are susceptible to. Furthermore, it will delve into the challenges of explainability and transparency in AI systems, emphasising the need for interpretable models to build user trust. This chapter will then proceed to analyse the future implications, exploring the potential consequences of AI-driven misinformation and disinformation detection systems on society, where ethical

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considerations must be prioritised, and the evolving legal frameworks necessary for effective data protection. By integrating insights from computer science, social sciences, law, and ethics, this chapter aims to reveal the path forward in safeguarding truth in the era of AI and ML.

5.1 Introduction

In an era dominated by information exchange, the prevalence of having more information has been seen as a common indication of trust in supporting the decision-making process of a source's credibility. Moreover, for many, the assumption is that information is a central part of the decision-making process and that having a higher quality will result in a superior outcome [49]. Unfortunately, when dealing with information, people must now consider the trustworthiness of the information presented. As information continues to play a prominent role in society, the danger of misinformation and disinformation threatens the credibility of information and poses a significant risk to the virtue of knowledge [42]. For instance, misinformation and disinformation disseminated through digital platforms can cause serious harm by undermining, for example, the integrity of election campaigns or public health messages [16]. Through mixing information that has been legitimately developed and distributed with intentionally (disinformation) or unintentionally (misinformation) inaccurate information, the credibility of the intended actual message may be lost in confusion.

In this chapter, we define misinformation as information that is false but has not been created with the direct intention of causing any harm, whilst disinformation is the creation of deliberately false information with the direct intention of causing harm to a person, organisation or social group [53]. As misinformation is often considered not to be purposely spread, it is most commonly disseminated through routes such as rumours and works of fiction [25] or parody and clickbait material [53]. Alternatively, with disinformation intentionally attempting to cause harm, it is often associated with the motivation to gain benefit, such as for personal or professional advantage (i.e. financial or political wealth) [16].

In recent years, it has been said that dissemination of misinformation and disinformation has increased rapidly due to the adoption of social media. As stated by Turcilo and Obrenovic [51], misinformation and disinformation are not new phenomena, but the widespread application and utilisation of social media have made this a more pressing issue. Another factor increasing the prevalence and awareness of misinformation and disinformation originated in response to the Russian disinformation campaigns. In response to these campaigns, the term 'fake news' was coined by the then-elected US president Donal Trump, who labelled disinformation as 'fake news' [16]. Since then, the term 'fake news' has seen widespread adoption, especially during the coronavirus disease of 2019 [40], where large quantities of inaccurate information were disseminated through social media sites, as demonstrated in Fig. 5.1, with over 62% of respondents of a global study conducted by

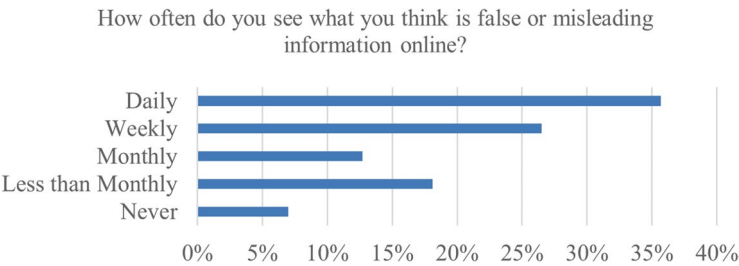


Fig. 5.1 Poynter [44]

Poynter [44] revealing that they believe to have encountered misinformation or disinformation online on a weekly basis.

As the widespread presence of misinformation and disinformation continues to contaminate credible and reliable sources of information, technologists must turn to new technologies to help limit the impact false information can cause. This chapter will explore how advancements in machine learning (ML) and artificial intelligence (AI) may offer a glimpse into restricting the dissemination of false information. Moreover, it will explore how cutting-edge detection techniques and content analysis tools can act as fact-checkers, ensuring the credibility of information.

5.2 The Rising Concerns of Misinformation and Disinformation

Documented in the official European Union report of 2018 [15], it was stated that misinformation and disinformation now influence at least two-thirds of citizens at least once a week. The report details how the widespread dissemination of false information has prompted users to think critically about the integrity of the information presented on their screens. However, some authors have claimed that the increase in exposure to misinformation and disinformation is predominantly down to social media. Due to its easy access, social media has become a platform allowing users to express ideas and publish news freely without any hindrance or control over the authentication of what is published [27]. However, due to this freedom and the vast number of users on the platforms, social media has become a hotspot for the spread of this false information.

The concerns surrounding misinformation and disinformation are highly disruptive and create a sense of uncertainty and doubt surrounding information, causing huge issues not only for society but also in business operations [2]. Interestingly, reported in the 2017 MIT Technology Review, Snow [48] stated that it was expected that by 2022, most individuals would be consuming more false information than valid and accurate information. As such, due to the rising concerns surrounding misinformation and disinformation, some authors have said AI and ML could help

manage the dissemination of accurate information by detecting false information [14]. However, in the same report documented by Gartner, it was expected that the creation of a 'counterfeit reality' through the dissemination of misinformation and disinformation would surpass the ability of AI to detect it [38]. Therefore, the ability of AI systems to detect misinformation and disinformation may fall behind the production and dissemination of the information.

Additionally, with the increase in the adoption of data-driven systems, it is no surprise that a counterfeit reality of misinformation may surpass AI detection tools. Moreover, there is unparalleled access to information due to the introduction and rise of digital communication platforms such as social media, online news portals, and other digital channels. This has changed the traditional ways of sharing information. There is an information overload due to the vast amount of data accessible via digital platforms. Although having so much information empowers people to see things from different angles, distinguishing between false and accurate information is complex, with potentially disastrous consequences for individuals and communities. This has also caused profound disruptions to democracies, societies, and personal opinions.

The ease of sharing information globally has helped spread information virally through culturally diverse and geographically distant parts of the world instantaneously by exposing societies to diverse ideologies. This has raised concerns about cultural values at the individual level more than at the country level [22].

People in the twenty-first century gather information, especially news, from the internet. This accessibility has disadvantages in addition to many other benefits. The most significant disadvantage is the dissemination of misinformation and disinformation. Democracies are currently battling the impact of this false information. As we have seen in some elections worldwide, public opinion can sway based on online information. The danger of this is that it could result in people making illogical decisions, which may have dangerous consequences in the long term.

A survey indicates that a growing number of Americans get their news via social media [20]. Furthermore, according to a different study, younger individuals are more likely than older people to rely on Internet sources to obtain news [36]. In light of this, it is fair to say that if the accuracy is not checked, a considerable number of people will access misinformation and disinformation, making them more vulnerable. Although social media companies attempt to slow the spread of misinformation by implementing fact-checking programs, the volume and speed at which information spreads online are challenging. This makes programmes such as fact-checking programs struggle to keep pace with the flow of information on the Internet, resulting in them not acting as effectively as they could be.

On the other hand, finding the ideal balance between the necessity to restrict harmful content and the right to freedom of expression is one of the most significant issues. Although everyone has the freedom to voice their thoughts, this right is not absolute, particularly when it comes to spreading misinformation and disinformation. Therefore, social media businesses have also taken action to identify and ban

the accounts that disseminate false information. Nevertheless, the question is whether those efforts are sufficient and adequate.

It is also the responsibility of social media users to identify and act against Fake News by reporting these items to the relevant parties and avoiding interaction with such information. In addition, promoting media literacy and critical thinking abilities among the public is essential to hold platforms accountable for spreading misinformation and disinformation. It is also imperative for policymakers to strengthen rules and enforcement protocols to prevent the dissemination of misinformation and disinformation on the internet.

Fundamentally, as information continues to firmly become a critical resource in modern technologically driven societies, the requirements for individuals and organisations to act upon misinformation and disinformation are essential. If fundamental steps are not taken, the integrity of digital information may be compromised beyond repair. It is, therefore, the technologist's role to incorporate future technologies to help battle and reduce the spread of misinformation and disinformation.

5.3 Application of AI and ML Detection Techniques

Manually detecting misinformation and disinformation is a time-consuming and labour-intensive task that is often completed on a case-by-case basis [23]. However, with the spread of false information (especially during the COVID-19 crisis), alternative methods of detection are required, explicitly applying artificial intelligence tools to detect false news automatically [47].

Artificial intelligence offers the ability to enable an AI-driven fact-checking system that can automatically compare information (such as contained within social media posts) against reliable sources to determine its accuracy [23]. The ability of these systems would enable a user to insert information that raises doubts and cross-reference it against information from sources that are deemed to be credible. According to certain authors, such as Nasir et al. [33], the application of AI techniques is particularly effective in exposing false information contained within social media platforms. However, other authors claim that the application of AI in the detection of disinformation requires a multifaceted approach that should also consider human verification [46]. Moreover, the use of AI is only one step of an intricate process in detecting misinformation, and human verification should also be applied to ensure the integrity of the information.

When detecting misinformation on social media platforms, it is important to remember that not all platforms are the same and that they all have unique features [11]. These features must be taken into account when developing detection algorithms.

5.3.1 *AI and ML Misinformation and Disinformation Algorithms*

Research on detection frameworks has seen considerable progress in recent years, with current ML and Deep Learning algorithms being adapted to tackle MI and DI online; the following is just a small number of algorithms aiming to improve detection.

5.3.1.1 Convolutional Neural Networks (CNN)

Yu et al. [55] propose the use of Convolutional Neural Networks (CNN). In addition to failing to establish meaningful interactions among features, the authors argue that traditional feature engineering techniques centred around Recurrent Neural Networks (RNN) are ill-suited to capturing the dynamic nature of social media and provide limited capability of early misinformation detection. By using CNN to extract important features and determine how they interact, the Convolutional Approach for Misinformation Identification (CAMI) overcomes these limitations and makes it possible to identify and detect misinformation more effectively and early on. CAMI's effectiveness in both tasks is validated by experimental results on large-scale datasets. Because of the quick development of social media, the paper emphasises the increasing importance of early misinformation detection. CAMI introduces paragraph vectors to represent microblog posts and utilises CNN to automatically extract key features from both misinformation and truthful information. CAMI's innovation lies in its capability to automatically extract significant local-global features, unveil high-level interactions, and flexibly identify key features dispersed within input sequences. Evaluation of large microblog datasets demonstrates CAMI's superiority over existing methods in both misinformation identification and early detection tasks.

5.3.1.2 Attention

Liu and Wu [26] presents research on the early detection of fake news on social media using convolutional and recurrent networks to classify the propagation path. The model combines recurrent and convolutional networks to capture user characteristics and focuses on classifying news propagation paths. According to experimental findings, the suggested model can accurately identify fake news within 5 minutes of its spread. The study emphasises the value of early detection as well as the shortcomings of current methods. When compared to baseline models, the suggested model detects fake news more effectively and efficiently. Future work surrounding this research will examine how user traits can be used to identify users who are likely to disseminate false information.

5.3.1.3 CNN + LSTM

Dhamani et al. [13] investigated the use of transfer learning and deep networks in combating misinformation. For text classification, character-level convolutional neural networks work well, enabling social media features and non-vernacular speech. For examining misinformation on social media, the CNN- Long Short-Term Memory (LSTM) model is a versatile tool. Transfer learning works well in natural language processing. The model can categorise emails and identify communications that are malicious. Moreover, the model can be used to identify and counteract misinformation and inauthentic behaviour. A semantic classifier with multiple labels and classes is the suggested approach. Bidirectional, dropout, max-pooling, and convolutional LSTM layers make up the model. Classifying content as intentionally neutral, intentionally beneficial, or intentionally harmful is crucial. Posts that are classified as harmful or damaging are accurately identified. Numerous domains can benefit from the CNN-LSTM classification approach. Large-scale disinformation campaigns can be understood by the model, and features for exploratory data analysis can be learned. Misspellings and unusual vocabulary usage are permitted by the character-level CNN. For sentence encoding and document classification, the model consists of two deep neural networks. After being trained on several datasets, the classifier obtains a test binary accuracy of 96.14 per cent.

5.3.1.4 BiLSTM

Zhang et al. [59] developed the FAEKDETECTOR model. When the researchers compared the model with comparable alternatives, they found that it has superior accuracy and credibility when deriving information from news articles. To identify fake news articles, creators, and topics, the model makes use of deep learning and heterogeneous network analysis techniques. The model takes into account information about the network structure and extracts explicit and latent features from text data. The model's exceptional ability to detect false news has been demonstrated through extensive testing on the PolitiFact dataset.

5.3.1.5 XFlag + LSTM

Chien, Yang, and Yu [10] propose an inventive explainable AI framework called XFlag, which makes use of the Situation Awareness-based agent Transparency (SAT) model to improve transparency in human-AI interaction, the Layer-wise Relevance Propagation (LRP) algorithm to explain the fake news detection model, and Long Short-Term Memory (LSTM) to identify fake news articles. The findings demonstrate how well the XFlag framework works to help users comprehend system operations and confirm the accuracy of the source. The study also looks at how user understandability, explainability, workload, and trust are affected by system transparency. The suggested XFlag framework raises transparency of AI models,

helps users determine the legitimacy of sources, and enhances model accuracy while facilitating human–AI interaction. The research emphasises the significance of providing end users with clear explanations and transparent information. The model attains noteworthy outcomes in accuracy and F1 score, comparable to state-of-the-art models trained on the Weibo dataset, according to some aspects of the authors’ conclusions, which claim to validate earlier research in this area. Chien acknowledges that the proposed XFlag framework has limitations in terms of the current dataset and the use of the aggregation approach in LRP. Future research will address these shortcomings by enhancing the framework’s functionality and applying it to various datasets. Additionally, user input will be gathered to enhance the framework.

5.4 Content Analysis

Zhang et al. [57] reported on a computational approach to the detection of misinformation in which a framework processes data in real time, which was built by a research group at Auburn University. The approach involves utilising a web crawler to pull news events from legitimate sources to build a dataset of events. Traditional detection techniques have utilised the use of LDA to build topic models based on keywords. However, this approach utilises Stanford’s CoreNLP to build a list of events from sentences within the news articles and categorise them into topics. It was found that this approach proved to have better accuracy and effectiveness when detecting fake news during testing compared to previous models such as FEND. The research group has expressed limitations in detecting fake news in real-time. The proposed model cannot handle evolving events or events where news outlets might disagree about the facts of an event, and in such cases, this might classify legitimate news as fake. Future work on this model would require the use of time-stamping events and purging outdated articles to improve accuracy.

5.5 Explainability and Interpretability

Explainable Artificial Intelligence (XAI) or Interpretability in the context of ML is an important consideration when designing an AI- or ML-based detection system, to ensure that the human user can trust the decision-making process of the models and bridge the gap between artificial and human intelligence.

The National Institute of Standards and Technology (NIST) [43] set out four main principles that XAI systems should address in order to foster confidence with the human user, no one of these principles can be used alone and for systems to truly gain user trust all must be used together and build upon the previous principles.

5.5.1 Explanation

Processes and/or outputs must be accompanied by evidence or reasoning that supports the decisions made by the system. Although it must be noted that NIST have not explicitly defined what types of evidence would be justifiable in this situation due to the large scope of use of AI and ML applications.

5.5.2 Meaningful

This is a measure of explanation; has the user understood the explanation that has been given? Miller [28] discusses how, as humans, we tailor our explanations of a topic to the individual receiving that information based on their knowledge of the topic, their cultural background, sex, and age. Building an XAI model based on existing human-generated explanations would improve the overall meaningfulness of explanations. Furthermore, Miller explains that research in social sciences has found that for an explanation to be meaningful, it must be built on simplicity, coherence, and generality.

5.5.3 Explanation Accuracy

Phillips et al. [43] explain that although the system's decision-making judgement might have been accurate, this is different from the accuracy of the explanation. Explanation accuracy expands on meaningfulness, and accuracy is dependent on whether the user has been able to understand the explanation given. An explanation for the decision of the system can be given accurately at a high level, but if the receiving user is not knowledgeable in the area, then the accuracy of the decisions made would be lost on the user. Papenmeier et al. [39] discovered that research suggests the only explanations that improved user trust and understanding were preceded by 'why' and 'why not', supporting the principle of Meaningful through keeping explanations simple to improve accuracy.

5.5.4 Knowledge Limits

Previous principles have assumed that a system has been able to accurately provide a response within its scope of application. An explainable application must also understand its own limitations and where an output does not fall within expected boundaries the system must be able to explain this to the user as to why a prediction

was not achievable at this time, this could be down to an unexpected input or a lack of credible evidence to make an informed prediction [43].

5.6 Deepfakes

It must be remembered that misinformation and disinformation are not just a text-based problem, but with the advances being made in the creation of AI-generated images, the issue of Deepfake generation must also be considered. Deepfakes are AI-generated images or videos that, to the human eye, appear to be genuine, with the most common form of deepfakes created through the manipulation of genuine human imagery [29]. Research by Ng [35] has shown that although users are sceptical when it comes to the trustworthiness of some online content, when a user already has a connection with an individual, such as a well-known celebrity or politician if the content appears to be in line with what they would expect the individual to discuss, their likelihood to perceive a deepfake as trustworthy increases. Although current research shows that users possess the knowledge to identify a deepfake, the increasing capabilities of AI-generated imagery and videos mean that deepfake detection algorithms cannot be overlooked. Furthermore, as more users engage with visual content via social media platforms, the dangers that deepfakes hold may be significantly intensified in the coming years.

5.7 Algorithmic Bias and Fairness

Neumann et al. [34] address the lack of investigation into fairness in automated misinformation detection systems. The research emphasises how important it is to have justice frameworks that cover every facet of equity in information settings. Additionally, it discusses the possible drawbacks for various stakeholders such as information subjects and evidence providers. It is stressed how crucial it is to recognise algorithmic mistakes and the different ways in which they affect different groups. The study recommends for the creation of technical countermeasures to disinformation.

Algorithms in newsfeed could potentially encourage the spread of misinformation and disinformation while also undermining the authority of genuine sources [30]. Users become isolated and have less variety in their news sources due to echo chambers and filter bubbles based on their browsing history. Algorithms that recommend news items require better guidelines and standards. Paid content and algorithmic bias can also impact newsfeed performance. Discrimination and the deliberate dissemination of false information can result from an algorithm's lack of transparency.

Cheng et al. [9] introduce the concept of Socially Responsible AI Algorithms (SRAs), where the need to understand the connections among different dimensions

of AI is highlighted. AI systems have obligations that fall within four categories: philanthropic, legal, ethical, and functional. The misuse of data and the absence of user consent are emphasised as problems within AI algorithms. SRAs should be centred around several important goals, including safety, fairness, transparency, and trust. However, fairness in AI is difficult to achieve because of its subjective and evolving nature. Within the context of SRAs, the research defines fairness within the decision-making process as something that is absent of prejudice or favouritism. Errors made by Stanford's COVID-19 vaccine allocation algorithm, which gave older employees priority over frontline workers, are used to highlight the need for transparency and fairness in AI algorithms.

5.8 Adversarial Attacks

Adversarial machine learning can be applied to amplify attacks and avoid detection [4]. Compared to its counterparts in computer vision, research on adversarial defence in natural language processing is still in its infancy [5].

Mosca et al. [31] propose an adversarial detector that uses logit variation interpretation. While this method has shown promise in identifying adversarial images, it has not yet been used in Natural Language Processing (NLP) adversarial defence. Current character-level defence techniques have proven successful in identifying syntax or spelling errors but have not proven effective against attacks that preserve language correctness. The technique uses machine learning models and feature attribution explanations to distinguish between adversarial and original samples and assign word relevance scores to an event. The adversarial detector receives the data from the pipeline, which gauges the model's response to inputs. Many detector architectures were examined, and word-level adversarial detection using the suggested approach performs better than the previous state of the art. Across various models, datasets, and attacks, the detector demonstrates generalisation capabilities.

Zhou et al. [60] propose that their Dirichlet Neighborhood Ensemble (DNE) defence method consistently outperforms existing methods. DNE uses a novel adversarial training algorithm to defend against strong attacks. Using the DNE method, words contained in input sentences are substituted with randomly selected points from a convex hull made up of the embeddings of the original word and its synonyms. Under the strongest attack, DNE achieves 77.2% accuracy, while other methods drop to 0.0%, 32%, 64.3%, and 8.1% accuracy. DNE substantially improves robust accuracy without sacrificing performance on clean data. The defence algorithm is broadly applicable and can be incorporated into any neural network. Adversarial-trained smooth classifiers outperform existing defences on various network architectures and datasets. Deep neural networks in NLP tasks are vulnerable to attacks, and word substitution-based attacks are discussed.

Yuan et al. [56] described bridging the gap between CV and NLP. They proposed a gradient-based textual adversarial attack framework. This paper introduces a textual adversarial attack framework for generating adversarial texts in Natural

Language Processing (NLP). The framework includes an effective attack method called Textual Projected Gradient Descent (T-PGD) that handles decision-based black-box attacks. The method transforms the problem of searching for adversarial samples in discrete text space into the continuous embedding space. Experimental results show that T-PGD outperforms other attack methods in terms of attack performance and adversarial sample quality. The method is evaluated on sentiment analysis, natural language inference, and news classification tasks. The success of the attack method is attributed to the more effective search process guided by gradient information. The quality of the adversarial samples generated by T-PGD increases with the text length. The method also yields higher Universal Sentence Encoder (USE) scores than baseline models, indicating more precise manipulation of adversarial samples. However, the grammatical performance of T-PGD is not the best on certain datasets. Yuan explains that the transferability of non-pretrained models to PLMs is still unknown, but this may not be a significant concern in practical scenarios when discussing possible improvements. The framework used to transfer adversarial attack methods from CV to NLP was instantiated with the PGD attack as an example, but it would be interesting to explore more attack methods and analyse their benefits for NLP. They propose that adversarial samples crafted by T-PGD have better quality than the baseline method in terms of grammar and fluency, according to human annotators. However, both BERT-Attack and T-PGD still have room for improvement in terms of grammatical correctness and fluency.

5.9 Case Studies

Misinformation and disinformation have grown to be serious issues not just in the United States but in other countries as well. After the Israeli government bombed Gaza and Hamas militants attacked Israel in 2023, verified users on Twitter published a vast amount of misinformation, recreating video game content, old videos, and altered photographs to promote partisan narratives [17].

People are already on edge after such incidents, and therefore, the dissemination of disinformation and misinformation has the potential to cause unnecessary stress, trauma, and psychological and emotional damage to people who have been directly impacted by the incident, as well as the other communities in the society. Additionally, if people panic, they may also act irrationally. Furthermore, it is typical for misinformation and disinformation to target specific racial, religious, or cultural groups, which can also lead to violence and increased isolation.

People occasionally use false and misleading information to fulfil their political agendas or incite social instability. If political and radical groups spread false information following an attack to pursue their personal agendas and goals, it will undoubtedly divide and polarise societies. On the other hand, the disinformation and misinformation may also cause law enforcement and investigators to incorrectly identify suspects, impeding efforts to stop similar attacks in the future and

giving suspects the opportunity to flee the scene without being prosecuted. This could then weaken public trust in political institutions and law enforcement.

It is also vital to consider that when false news spreads faster than wildfire, it can divert resources away from where they are needed most. Emergency responders may have to respond to false alarms, and it may waste time and resources on a threat that does not exist. Because of that, the real victims may end up not getting the assistance they require. Therefore, it is crucial to filter information before it is made public for all of the previously listed reasons. Unfortunately, misinformation and disinformation have already influenced many aspects of society, including the following instances.

5.9.1 2016 Presidential Election

During the 2016 US presidential election, fake news predominated on social media more than real news (Center for Information Technology & Society, ND), which raises questions regarding the integrity of democratic processes. The average US adult encountered one or more fake news articles during the 2016 election period, with higher exposure to pro-Trump content [3]. Guess et al. [21] found that older Americans were more likely to spread fake news by up to four times as much as those from within the younger demographic and suggested that research indicates that memory decline has an impact on their susceptibility to sharing of misinformation, although acknowledges that further research is required in this field. Despite efforts at the time to detect false news, it remained in circulation since this more significant effort has been made to combat the spread of misinformation and disinformation.

Another prominent example of misinformation in presidential elections is that of Cambridge Analytica. In this case, Cambridge Analytica created psychological profiles of social media users globally using data from social media. They were able to predict people's political leaning by obtaining Facebook user data without authorisation (Center for Information Technology & Society, ND). Subsequently, a deliberate effort was initiated through the production of Facebook political advertisements and memes with the aim of undermining Hillary Clinton's presidential campaign and influencing Americans' views in support of Trump (Center for Information Technology & Society, ND). The content that was produced was inaccurate, aggressive, and provocative messages.

5.9.2 2013 Boston Marathon Bombing

Within 45 minutes of the Boston Marathon bombings, misinformation about the event started to spread [50]. Three prominent rumours were identified, including false claims about a girl dying in the marathon. A Tweet circulated of an 8-year-old

dying in the bombing. Within 4 minutes of this Tweet, a false image of a girl had been added to aid in the spread of this false news. By only 33 retweets, the misinformation had started spreading on other platforms. The second speculation centred around the involvement of Navy Seals or Blackwater. Images of dark backpacks used in the bombing were released by the FBI. This resulted in a vigilante movement of one Reddit user to share images of men wearing similar backpacks in the area. One such image shared showed two men, one with a Navy Seals emblem, increasing speculation that the government were involved. The final rumour was the misidentification of Sunil Tripathi as a suspect: a university student who had gone missing some weeks before the bombing was falsely identified by a fellow student following the release of the police images of the suspects. This rumour gained less traction, though, following the release of the suspect's names by the FBI. The research found that due to the context of these rumours and the volume of misinformation, corrections were unable to slow the spread, with a misinformation-to-correction ratio of 44:1 for rumour one, 18:1 for rumour two, and 5:1 for rumour three.

5.9.3 Health Misinformation

The COVID-19 pandemic is another instance where, despite social media's potential benefits, fake news is mainly disseminated through it, making it harder for consumers to know what is correct and what is not. A significant amount of this false information went on to exacerbate phobias and anxiety. As a result of this, there were some odd patterns in the purchases of personal items such as toilet rolls [24] and the buying and using of drugs without authorised medical prescriptions [6]. These behavioural shifts may be detrimental and may have resulted in self-harm instances thus signifying the seriousness of disinformation. A number of reports went on to highlight real-world incidents, such as after hearing on the news that chloroquine could be able to treat COVID-19, a resident of the United States took it and passed away [6]. Another incident was identified in that the Iranian government disclosed that individuals had died from alcohol poisoning after drinking alcohol, believing that consuming illicit alcohol could protect them from the illness [1]. A study of more than 1700 US adults discovered that individuals, particularly those who rely more on intuition or have little scientific background, spread more false information about COVID-19 on social media due to a disregard for accuracy. A subtle reminder to verify information before reposting was found to make people less likely to spread misleading information [41].

Conspiracy theories are not new to a pandemic, and COVID-19 is not the first. Conspiracy theories about the Zika virus, Ebola, and possibly AIDS were also widely circulated [6]. The World Health Organisation alerted the world community to a developing 'infodemic' on social media in the early stages of the pandemic [54]. A report suggests that the WHO claims that this 'infodemic' is bringing about false medical advice, psychological panic, and economic disruption [32].

5.9.4 *Mass Shootings*

There was an incident where, after a major shooting in the USA, Conspiracy theorists were incentivised to post materials that were believed to have originated from Russian agents, political campaign operatives, and other fake news sources. These sources claimed that the shootings were staged to gain public support for gun control laws [8]. The story looked fake when they later examined the hashtags from tweets, web pages, and comments. Following the death of George Floyd in 2020, there was a national uproar in the United States, and racial and political divisions widened in the United States as a result of the widespread spread of fake news on social media, online rumours, and photoshop-manipulated photos [7]. In another horrific mass shooting that happened in a Maine restaurant and bowling alley towards the end of 2023 in the United States, false information about the perpetrator and his capture flooded social media platforms, including Facebook, Twitter, and TikTok [17].

The high levels of misinformation and false news pose a new major challenge for society. The extent of this challenge can have widespread effects, from shaping political procedures to instilling fear in the population. Events like the 2016 US presidential election, the Boston Marathon bombing, health misinformation during the COVID-19 pandemic, and mass shootings illustrate the gravity of this problem. Reducing the impact of false information and misleading content necessitates people to spend time verifying the reliability of sources. However, additional effort is needed to encourage responsible behaviour on social media platforms and improve media literacy.

5.10 Legal and Ethical Considerations

Due to the complex nature of detecting misinformation, ethical considerations need to be made surrounding the utilisation of AI for detection purposes. The use of AI for the detection of misinformation and disinformation raises issues relating to privacy, accountability, and transparency [46]. In order to successfully apply AI techniques to detect false information, Santos [46] states that we must be guided by ethical principles and fundamental human values.

Misinformation and disinformation are widespread problems in the current digital era, and they pose serious problems for people, governments, and societies everywhere. Therefore, investigating the legal and ethical challenges involved in addressing misinformation and disinformation is of paramount importance. However, regulating misinformation and disinformation is difficult, as a thin line separates upholding freedom of expression from reducing the damaging effects of misleading information. The ongoing dissemination of fake material online seems to have been facilitated by the absence of a thorough regulatory structure. It would

take time to develop a robust structure, but it is time for the policymakers to formulate policies that work while upholding democratic values.

The anonymity of the source is another difficulty for policymakers and government officials dealing with misinformation and disinformation. This implies that people who disseminate false information occasionally attempt to conceal their identity by creating fake profiles. This damages the source's credibility and sense of trustworthiness. However, secretly disseminating false information does not exclude a person from governmental and judicial scrutiny. Regardless of the anonymity, rules and regulations are in place in many jurisdictions to handle the spread of fake news. If anonymous actors break any laws related to incitement, fraud, or defamation, they could be held legally liable. In addition, platforms may also use reporting tools and AI and ML algorithms to recognise and eliminate damage. This might also assist authorities and platform owners in combating misinformation and disinformation. In reality, even though anonymity provides a cloak of invisibility for spreading misinformation and disinformation online, anonymity does not shield content from removal or platform policies. On the other hand, if people can be extra cautious when accessing content from unidentified or anonymous sources, it would make it more difficult for anonymous sources to spread misleading information.

The capacity of criminals to intentionally fabricate and forge data in Big Data is also another concern [58]. This false data will unavoidably provide inaccurate outcomes. Sometimes, people may also fabricate data to create illusions that work in their favour and mislead others [58]. Some websites contain false comments and ratings, and users can easily be lured into buying goods and services based on those fake comments and ratings. On the other hand, it is also possible for people to fabricate datasets, which could cause doubt surrounding the authenticity and credibility of the open-source databases that researchers can access.

According to the International Human Rights Law, governments must guarantee adequate access to truthful information [51]. This type of access encompasses the freedom to look for and obtain information. In light of this, governments must support and safeguard unrestricted access to accurate and reliable information.

A number of nations have put laws and regulations in place to deal with misinformation and disinformation, especially in relation to social media and Internet platforms. These laws frequently imprison people and issue fines for those who disseminate false information. However, the effectiveness of these laws depends on the acceptance of existing legal frameworks. The online safety Bill in the United Kingdom addresses the issues regarding misinformation and disinformation [19]. The measure intends to hold digital companies accountable for the content posted on their platforms and has included new criminal offences related to online activities [45]. This will encourage authorities and organisations to keep an eye out for and halt the spread of false information. The president of Kenya signed the Computer and Cybercrimes Act into law in 2018 [37]. The act makes it illegal to broadcast fake news; if someone is found guilty, they will receive a two-year prison sentence, fines of up to nearly \$50,000, or both. The law further states that it is illegal to publish false, misleading, or fraudulent material knowingly or to mislead people knowingly. The Penal Code of Sri Lanka also contains general provisions that address

various types of false statements [18]. They are meant to enable action against those disseminating false information, particularly remarks that could jeopardise national security and encourage intergroup conflict.

Legislation against spreading false information through electronic devices was passed in Tunisia, although many people are against it because they believe it will limit freedom of expression. Article 24 in the legislation stipulates that anyone found to be disseminating false information and rumours online faces a maximum sentence of five years in prison and a fine of up to \$15,000 [12]. Most importantly, that sentence doubles if the fake news is directed at a state official. Critics argue that the new legislation has given lawmakers an easy way to punish citizens as it does not specify what kind of information or rumours are prohibited [12]. Furthermore, critics claim that some clauses will let authorities check computers or telecommunication devices, seize equipment, and intercept data if authorities believe there is a violation of the new legislation [12].

5.11 Discussion

This chapter has explored how misinformation and disinformation on social media are pressing concerns today. Threat actors are using advanced technology to evade detection, making it harder to combat this problem with machine learning and artificial intelligence alone. Collaboration among various stakeholders is vital to address this challenge. Government institutions can enact legislation to regulate the spread of false information and penalise those who deliberately spread misinformation. They can also fund research into new technologies and methodologies to combat this issue.

Academia can contribute by studying the patterns and impacts of misinformation and disinformation, developing new strategies and tools to detect and counter them, and promoting educational initiatives to increase public awareness about the dangers of false information.

Social media platforms, as primary conduits of information in the digital age, have a significant role to play. They must promote awareness about the risks of misinformation and disinformation through user education programs and implement early warning systems that notify users of potential false information on their platforms.

Equipping users with knowledge to identify false information is critical. By leveraging AI and ML technologies to support users, they can become active participants in safeguarding themselves and society against these threats. This could involve developing user-friendly tools that help identify unverified information or suspicious sources.

The successful implementation of early warning systems and education initiatives has the potential to significantly reduce the impact of misinformation and disinformation on society, foster a more informed society, reduce polarisation caused by false narratives, and promote a healthier discourse in the digital public sphere.

5.12 Conclusion

In an information age, the threat of misinformation and disinformation poses significant risks to the credibility and integrity of information online. As this chapter has discussed, false information presented online can have significant implications that may result in mental or physical harm. The chapter has highlighted numerous case studies in which the dissemination of false information has resulted in varying degrees of impact. Primarily, this chapter has identified methods in which AI and ML may be utilised to identify and mitigate the risks posed by false information effectively. However, in identifying the potential of AI and ML, numerous technological restrictions and ethical dilemmas have been raised. However, the threat of misinformation and disinformation will continue to impair a reader's judgement on the integrity of information until companies (such as social media sites) adopt reforming changes to their operations. Moreover, implementing AI and ML directly into their operations to detect, confirm, and dispose of misinformation and disinformation in a legally abiding manner.

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